MILITARY SPECIFICATION

BOUNDARY LAYER CONTROL SYSTEM; PRESSURE

This specification has been approved
by the Bureau of Naval Weapons,
Department of the Navy.

1. SCOPE

1.1 This specification covers pressure boundary layer control systems of turbo-jet and turbo-prop aircraft for increasing the aerodynamic lift of the wings and flaps during approach, landing, and takeoff.

2. APPLICABLE DOCUMENTS

2.1 Government documents normally furnished - The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-D-8708	Demonstration Requirements for Airplanes
MIL-B-81365	Bleed Air Systems, General Specification for

(When requesting applicable documents, refer to both title and number. Copies of unclassified documents may be obtained from the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Requests for copies of classified documents should be addressed to the Naval Supply Depot, via the cognizant Government inspector.)

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3. REQUIREMENTS

3.1 Prototype component inspection - The boundary layer control system furnished under this specification shall be a product which has been inspected and has passed the prototype component inspection specified herein.

3.2 General - The requirements of MIL-B-81365 shall apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict this specification shall govern.

3.3 Design and construction -

3.3.1 Multi-engine aircraft - For multi-engine aircraft, provisions shall be made to prevent loss of boundary layer control air to a non-operating engine.

3.3.2 Manual operation - The boundary layer control system shall be operated by a single control in each cockpit. The control shall be the same control that actuates the wing flaps.

3.3.3 Controls - Controls shall be provided for symmetrical operation of the boundary layer control system. The controls shall be designed to preclude asymmetric flow over the wings and wing flaps.

3.3.4 Signal lights - Signal lights shall be provided on the instrument panel that shall indicate any abnormal condition in the boundary layer control system. The switches, wires, and other parts of the warning system shall be protected against deterioration from hot air. In dual flight-control aircraft separate signal lights shall be provided for each pilot unless a single set of signal lights is clearly visible to each pilot.

3.4 Performance - The boundary layer control system and components shall meet the performance requirements of MIL-B-81365.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection - The inspection of the boundary layer control system shall be classified as follows:

(a) Prototype component inspection

(b) Quality conformance inspection

4.3 Prototype component inspection - Prototype component inspection on the boundary layer control system shall consist of the following and in the order listed:

Sample 1 (Components and ducting)

(a) Proof pressure	(4.5.2)
(b) Endurance	(4.5.3)
(c) Flow resonance	(4.5.4)
(d) Burst pressure	(4.5.5)

Sample 2 (Bellows, valves, and ducting)

(a) Vibration	(4.5.6)

4.3.1 Sampling for prototype component inspection - Unless otherwise specified, as soon as practicable after award of a contract or order, the contractor shall furnish prototype samples for inspection to determine conformance with this specification. Prototype samples shall consist of two complete boundary layer control systems on contract or order. The samples shall be identified with the manufacturer's part number and contract number and shall be inspected as specified herein and in accordance with the contract or order (see 6.2).

4.3.1.1 A sample shall consist of all components necessary to assemble a complete system.

4.4 Quality conformance inspection - The quality conformance inspection shall consist of individual inspection.

4.4.1 Individual inspection - Each boundary layer control system on the contract or order shall be subjected to the following inspection in the order listed. Any system component containing a defect or failing to pass any of these

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tests shall be rejected and replaced. After replacement of the component, the system shall be completely reinspected in the order listed.

(a) Examination (4.5.1)

(b) Ground operation (4.5.7)

(c) Flight (4.5.8)

4.5 Inspection methods -

4.5.1 Examination - The system shall be thoroughly examined to determine conformance with this specification and applicable drawings with respect to all the requirements not covered by tests.

4.5.2 Proof pressure - The boundary layer control system shall be subjected to and shall pass the proof pressure test specified in MIL-B-

4.5.3 Endurance - The boundary layer control system shall be subjected to and shall pass the endurance test specified in MIL-B-81365.

4.5.4 Flow resonance - The boundary layer control system shall be subjected to and shall pass the flow resonance test specified in MIL-B-81365.

4.5.5 Burst pressure - The boundary layer control system shall be subjected to and shall pass the burst pressure test specified in MIL-B-81365.

4.5.6 Vibration - The boundary layer control system shall be subjected to and shall pass the vibration test specified in MIL-B- 81365.

4.5.7 Ground operation - Ground operation tests of the system shall be conducted with the aircraft at rest using full operating pressure to demonstrate system control, general security, structural integrity, and safety of the system for flight. The boundary layer control system shall show no evidence of leaks.

4.5.8 Flight - Tests shall be conducted on the boundary layer control system as part of the flight test program to evaluate the effect of the boundary layer control system on the performance and handling characteristics of the aircraft. The handling characteristics of the aircraft at gross weights and centers-of-gravity as specified in MIL-D-8708 during takeoffs, landings, cruise, and at high-speed conditions shall be demonstrated. Performance of the aircraft shall not be adversely affected by the operation of the boundary layer control system. The flight tests shall

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satisfactorily demonstrate the boundary layer control operation. The following flight test shall be performed at minimum safe altitude: With the aircraft trimmed at 1.2 times the stall velocity in approach configuration at level flight, the throttle shall be cut to idle power in less than 1/2 second, and it shall be shown that no unsafe characteristics exist due to the change in boundary layer control effectiveness.

5. PREPARATION FOR DELIVERY (Not applicable)

6. NOTES

6.1 Intended use - The boundary layer control system is intended for use in turbo-jet and turbo-prop aircraft to increase the aerodynamic lift of the wings and flaps during approach, landing, and takeoff.

6.2 Information for contracting officer - Contracts or orders should specify the following:

(a) Title, number, and date of this specification.

(b) Items of data required (see 6.3).

(c) Engineering change procedure in accordance with BUWEPS-INST 4275.2A, enclosure (1).

(d) Boundary layer control systems subjected to tests other than those listed under quality conformance inspection are not to be considered or shipped as part of the contract or order.

6.2.1 Contracts or orders should specify the following provisions for prototype component inspection.

6.2.1.1 Whether prototype component inspection is required. When a contractor is in continuous production of the items from contract to contract, consideration should be given to waive the prototype component inspection. If prototype component inspection is required, indicate:

(a) where the prototype component inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

(b) that the approval of prototype samples or the waiving of the prototype component inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

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6.3 Data - Data information shall be in accordance with MIL-B-81365.

6.3.1 Prototype data - When prototype samples are submitted (see 6.2) they should be accompanied by one complete set of manufacturer's drawings, the contractor's model specification, a parts list, and a complete test report showing the results of the contractor's tests. The test report shall include the following:

(a) Report of all tests, graphically presented when possible, together with a detailed statement indicating compliance or extent of noncompliance with all requirements of this specification, referring specifically to paragraph numbers. Wherever a requirement is considered to be not applicable, the report should so state.

(b) Summary of endurance and vibration tests.

(c) Diagrams of test setups.

(d) Reproducible outline and description of tests and test conditions. Where tests specified in this specification are not considered applicable, the reason and the substituted test shall be clearly described.

(e) Copies of test log sheets.

(f) Photographs when available.

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